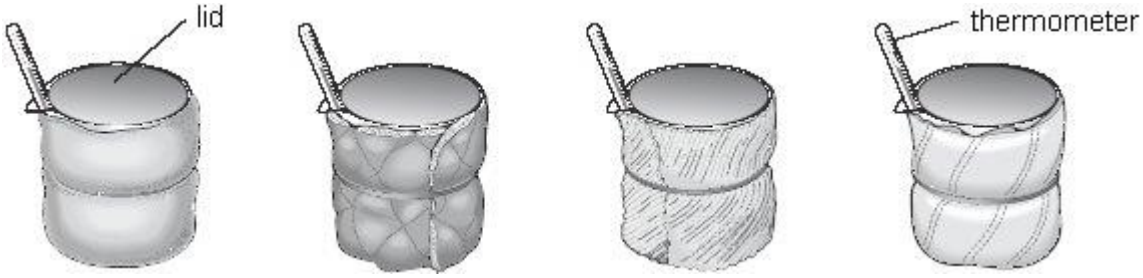
Q1.

A company has made a new material called 'Wellwarm'. They want to use 'Wellwarm' to make coats.

(a) A scientist tested 'Wellwarm' to see how well it insulated a beaker of hot water.
She tested 'Wellwarm' and three other materials as shown below.



material A **material B** **material C** **material D**

She wrapped each beaker in a different material.
She recorded the temperature at the start and 20 minutes later.

(i) What was the independent variable that the scientist **changed**?

.....

1 mark

(ii) What was the dependent variable that the scientist **measured** during the investigation?

.....

1 mark

(b) The results of the investigation are shown below.

time (minutes)	temperature of water (°C) wrapped in			
	material A	material B	material C	material D
0	60	60	60	60
20	34	40	38	36

(i) The scientist said that the 'Wellwarm' material is the best insulator.
Which material was 'Wellwarm'? Use the results to help you.
Tick the correct box.

A ☐ B ☐ C ☐ D ☐

1 mark

(ii) Use the evidence in the results table to explain your choice.

.....

.....

1 mark

(c) The company made a coat from each of the four materials they tested.



A person tested the different coats by wearing each one in a cold room.
He measured the temperature inside each coat for 30 minutes.

Write down two **other** variables that should be controlled to make this a fair test.

1. .....

1 mark

2. .....

1 mark

(d) Write down one thing the scientists should do to make sure the person testing the coats is safe.

.....

1 mark

(e) Suggest **one** advantage of using a temperature sensor and data logger instead of a thermometer in this experiment.

.....

.....

1 mark
maximum 8 marks

Q1.

(a) (i) (the type of) material

accept 'fabric'

1 (L5)

(ii) the temperature after 20 minutes

accept 'temperature'
'temperature at start' is insufficient

accept 'temperature change'

accept 'how hot the water is'
'heat' is insufficient

1 (L5)

(b) (i) B ✓

if more than one box is ticked, award no mark

1 (L5)

(ii) beaker B was warmer (at the end than the others)

accept the converse
accept 'the temperature was higher'
accept 'it dropped the least'
accept 'it lost the least heat'

'B was 40°C at the end' is insufficient

1 (L5)

(c) any **two** from

- temperature in cold room

  'temperature' is insufficient
  'the person' or 'the time' is insufficient

  accept 'his temperature before he went in'

- style or size of coat

  accept 'amount of material'

- the clothes he wears for each test

  accept '(same) underclothes'

- level of activity

  accept 'he had eaten the same food before each test'

2 (L5)

(d) • do not let the volunteer's body temperature go down too far

accept 'do not let him or the room get too cold'
accept 'have a doctor nearby'
accept 'monitor or check him'
accept 'measure his heart rate or breathing rate'
accept 'wear gloves' or 'keep his head warm'
accept 'make sure the volunteer is healthy'
or 'is not allergic to the material'

1 (L5)

(e) any **one** from

- monitor the temperature from outside the room

  accept 'remote sensing'
  accept 'you have to open the coat to read the thermometer'

- it gives you a continuous record (of the temperature)

  accept 'you do not have to write down the results'

- it is more accurate or precise

  accept 'it eliminates human error'
  'accurate' is insufficient

  accept 'the experiment is more reliable'
  'more reliable' is insufficient

1 (L5)

[8]